Supplementary Material for "Is This a Time for Irony? Estimating Ironic-Response Affordance in Dialogues"

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Abstract

This document contains the supplementary material for the paper. It reports the complete dialogue-evaluation protocol and annotation guidelines, including the scoring weights and Latin-square allocation, together with the full experimental configuration. Dataset documentation, sample instances, post-editing details, and prompts are distributed separately with the project repository.

I. Dialogue Evaluation

To evaluate CANTRIP's dialogues, we adopt CheckEval[1] as our evaluation framework for several key reasons. First, CheckEval provides a reliable, multi-dimensional and explainable approach by decomposing evaluation into fine-grained Boolean questions across multiple quality dimensions, making evaluation decisions more transparent and reducing ambiguity compared to holistic scoring methods. Second, its structured binary response format constrains the answer space, which helps LLMs generate more consistent and reliable judgments with reduced variability. Third, CheckEval has demonstrated strong correlation with human evaluation while showing reduced sensitivity to evaluator model choice, leading to more stable results across different LLM judges. Finally, its dimension-based decomposition into naturalness, coherence, engagingness, and groundedness aligns well with our need to assess whether generated dialogue turns integrate smoothly with the original conversational context.

Following [1], we first define the overall features of our automated evaluation strategy. The goal of the evaluation is to assess the extent to which the generated dialogues are relevant, coherent, and fluent w.r.t. the original context. The assessment is pointwise (each instance is evaluated separately) and reference-free (since we do not have a gold reference, each instance is examined w.r.t. quality criteria). Our setting encompasses a single LLM system first validated on a dataset subset against human judgment. The evaluation criteria include CheckEval's dimensions [2] detailed in the following and task-specific metrics related to our downstream application (Ironic-Response Affordance Estimation). The output of the evaluation consists of a numerical score that is self-explanatory due to the nature of the CheckEval's binary questions.

A. Annotation Guidelines

CheckEval [2] provides four main evaluation dimensions for Topical-Chat [3], a benchmark to evaluate dialogue systems that is the closest, most relevant application for the dialogue generation conducted in CANTRIP. Each dimension is broken down into subdimensions. For example, the dimension **Coherence** comprises *Logical Flow, Relevance,* and *Continuity*. Each subdimension is assessed through one or more precise questions, to which the annotator (human or model, depending on the stage of our framework) has to reply in a structured way (Yes, No). We report in the following the 16 adapted CheckEval questions that we adopt to evaluate the generation of preceding turns starting from an original one:

a) Coherence:: the quality of a generated turn that allows it to fit logically and meaningfully within the following context.

- *Logical Flow*
  - Does the generated turn logically precede the original turn, maintaining a clear flow of ideas?
- *Relevance*
  - Is the generated turn directly relevant to the content and context of the dialogue?
- *Continuity*
  - Does the generated turn stay consistent with the topic discussed in the dialogue?
  - Does the generated turn integrate smoothly with the ongoing conversation, ensuring a coherent progression?

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[1]https://github.com/yukyunglee/CheckEval

b) Engagingness:: the extent to which a generated turn captures and maintains a reader's interest.

- *Informative*
  - Does the generated turn add meaningful value to the conversation?
- *Emotional Engagement*
  - Is the generated turn friendly, polite, and empathetic?
- *Interest Level*
  - Does the generated turn contribute to a more engaging and interesting conversation?

c) Groundedness:: the accuracy and relevance of a generated turn in relation to the factual information it is based on.

- *Relevance*
  - Does the generated turn provide new information while maintaining the flow of the conversation?
  - Does it effectively utilize the key information that has been mentioned in the conversation?
- *Consistency*
  - Does the generated turn remain consistent with the original turn?
  - Does it avoid contradicting information provided later?

d) Naturalness:: the degree to which a generated turn resembles something a real person would naturally say.

- *Avoid repetition*
  - Does the generated turn avoid unnecessary repetition of the same content between sentences?
- *Context relevance*
  - Are the sentences used naturally within the context?
- *Clarity*
  - Is the overall message clear and easy to understand?
- *Word choice and tone*
  - Is the tone consistent throughout?
  - Is the text free of major grammatical errors?

B. Additional Dialogue Evaluation Details

Human annotators answered 16 binary questions covering the four evaluation dimensions described in I-A.

The human assessment of the generated dialogue was performed using a Latin square design [4]. The annotators independently evaluated the four experimental conditions, corresponding to the combinations of two LLMs and two different numbers of turns. Conditions were ranked using a weighted instance-level quality score computed from human judgments. In particular, following [2], which suggests that dimension-level scores can be combined using weights, we aggregated the binary annotations by first computing dimension-level averages and then applying weights reflecting their relative importance for dialogue quality in our setting. We assigned weights of $0.3/0.1/0.3/0.3$ to *Coherence, Engagingness, Groundedness,* and *Naturalness*, respectively, giving less weight to *Engagingness* as it is less critical for our purposes (more relevant in tasks such as question answering). Final scores for each configuration were obtained by averaging the weighted scores across all evaluated instances and annotators.

II. Experimental Details

All experiments were conducted on a GigaByte G242-P32-QZ server featuring an Ampere Altra Q80-30 CPU (80-core Arm Neoverse M1, 3.0 GHz, aarch64), 512 GB of RAM, two NVIDIA A100 GPUs and two NVIDIA BlueField-2 DPUs.

a) Models.: We evaluate a set of transformer-based models selected to represent different model sizes and training configurations (Table I). All models are used in their publicly available pre-trained or instruction-tuned versions without additional fine-tuning, unless otherwise specified. Table II shows the hyperparameters used during the fine-tuning.

TABLE I
TRANSFORMER-BASED MODELS USED IN THIS WORK, TOGETHER WITH THE NAMING CONVENTION ADOPTED THROUGHOUT THE PAPER.

Model	**Referred to as**
Llama-3.1-8B-Instruct	Llama3.1–8B
gemma-2-9b-it	Gemma2–9B
Gemma4–E4B-it	Gemma4–E4B
Qwen2.5-7B–Instruct	Qwen2.5-7B
Qwen3.5-9B	Qwen3.5-9B
Ministral-3-8B-Instruct-2512	Ministral3–8B
xlm-roberta-large	XLM–R
EuroBERT-210M	EuroBERT

TABLE II
HYPERPARAMETERS USED FOR TRAINING THE ENCODER MODELS REPORTED IN THE MAIN PAPER.

Parameter	**Value**
Training arguments	
Number of epochs	10
Batch size	32
Learning rate	2×10^{-5}
Weight decay	0.01
Warmup ratio	0.1
Focal loss parameters	
Focusing parameter (γ)	0.5
Weighting factor (α)	0.2

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